FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 13

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Recap *of previous lecture*

MATRIX



Topics to be covered

"DETERMINANTS"



⊛ only sq. Matrices have Determinants
$m=n$

eg Evaluate $|A|$ for

(i) $\begin{vmatrix} 2 & -3 \\ 4 & 1 \end{vmatrix}$ (ii) $\begin{vmatrix} 2 & -3 & 1 \\ 4 & 1 & -5 \end{vmatrix}$ = senseless

$= (2)(1) - (4)(-3)$

$= 2 + 12 = 14$

① $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{22}$

eg $\begin{vmatrix} 2 & 4 \\ 5 & -3 \end{vmatrix} = ? = (2)(-3) - (5)(4)$

$= -6 - 20 = -26$

$4 = \begin{vmatrix} 1 & 0 \\ 5 & 4 \end{vmatrix}, \begin{vmatrix} 2 & 2 \\ -1 & 1 \end{vmatrix}, \begin{vmatrix} 0 & 4 \\ 1 & -1 \end{vmatrix}, \ldots \infty$

$4 = \begin{vmatrix} 2 & 2 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}, \begin{vmatrix} -2 & 2 & 0 \\ +1 & 1 & 0 \\ 0 & 0 & -1 \end{vmatrix}, \ldots - \infty$

eg $\begin{vmatrix} 2+3i & -i \\ i & 2-3i \end{vmatrix} = ?$

$= (2+3i)(2-3i) - (i)(-i)$

$= 2^2 - (3i)^2 + i^2$

$= 4 - 9i^2 + i^2 = 4 - 9(-1) + (-1)$

$= 4 + 9 - 1 = 12$

Method of Calculating Det of order (3×3) →

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = ? = +a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

If we want to Expand along R_2 then

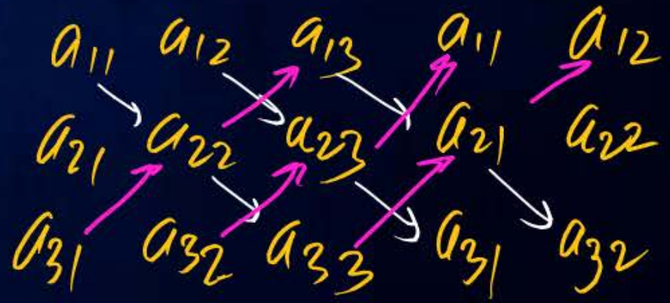
$$|A| = -a_{21}\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{22}\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} - a_{23}\begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$$

Sign Convention:

$$\begin{vmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{vmatrix}, \quad \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$$

Shortcut

SARRUS Method:



So $|A| = L - U$

Q

Evaluate $|A| = \begin{vmatrix} 3 & -2 & 1 \\ 0 & 4 & 5 \\ 1 & 3 & -4 \end{vmatrix} = ?$

$= +(3)\begin{vmatrix} 4 & 5 \\ 3 & -4 \end{vmatrix} - (-2)\begin{vmatrix} 0 & 5 \\ 1 & -4 \end{vmatrix} + 1\begin{vmatrix} 0 & 4 \\ 1 & 3 \end{vmatrix}$

$= 3[-16-(15)] + 2[0-(5)] + 1[0-(4)]$

$= -93 - 10 - 4 = \boxed{-107}$

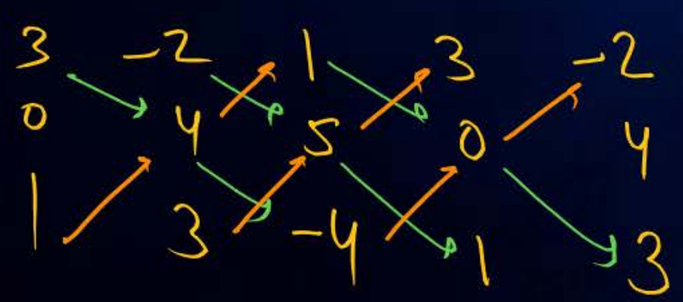
(M-II) Expanding along C_1:

$\begin{vmatrix} 3 & -2 & 1 \\ 0 & 4 & 5 \\ 1 & 3 & -4 \end{vmatrix} = +3\begin{vmatrix} 4 & 5 \\ 3 & -4 \end{vmatrix} - (0)\begin{vmatrix} -2 & 1 \\ 3 & -4 \end{vmatrix} + 1\begin{vmatrix} -2 & 1 \\ 4 & 5 \end{vmatrix}$

$= 3[-16-(15)] - 0 + 1[-10-(4)]$

$= -93 - 0 - 14 = \boxed{-107}$

(M-III)



$|A| = (-48-10+0) - (4+45+0)$

$= -58 - 49 = \boxed{-107}$

(Pg) Evaluate $|A| = \begin{vmatrix} 3 & -2 & 4 \\ 1 & 2 & 1 \\ 0 & 1 & -1 \end{vmatrix} = ?$ by using all three methods

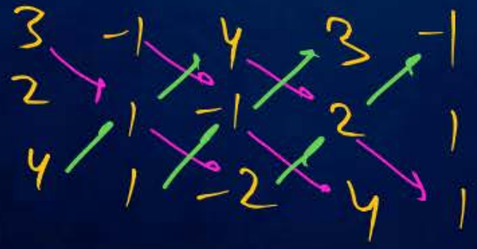
Ans = −7

Q. $|A| = \begin{vmatrix} 3 & -1 & 0 & 4 \\ 2 & 1 & 4 & -1 \\ -1 & 0 & 3 & 0 \\ 4 & 1 & 0 & -2 \end{vmatrix} = ? = +(-1)\begin{vmatrix} -1 & 0 & 4 \\ 1 & 4 & -1 \\ 1 & 0 & -2 \end{vmatrix} - (0)\,| \, ? \, | + (3)\begin{vmatrix} 3 & -1 & 4 \\ 2 & 1 & -1 \\ 4 & 1 & -2 \end{vmatrix} - (0)\,| \, ? \, |$

$= (-1)\left[-0 + 4\begin{vmatrix} -1 & 4 \\ 1 & -2 \end{vmatrix} + 0\right] + 3(?) - 0$

Try to Expand along R_3

$\begin{vmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{vmatrix}$



$\Rightarrow Det = (-6+4+8) - (16-3+4) = -11$

$|A| = -1\left[4(2-4)\right] + 3(-11)$

$= 8 - 33 = -25$

Properties of Determinants :-

(1) We can Expand along any Row or along any Column But it is advisable to expand along that Row or along that Column which has max number of zeros.

(2) $\begin{vmatrix} a & b & 0 & d \\ e & f & 0 & h \\ \alpha & \beta & 0 & \delta \\ x & y & 0 & w \end{vmatrix} = 0$

i.e if all the elements in a Row or in a Column are zero then it's Det = 0

(3) $\begin{vmatrix} a & b & c & a \\ e & f & g & e \\ \alpha & \beta & \gamma & \alpha \\ x & y & z & x \end{vmatrix} = 0$

if any two Rows or any two Columns are identical then it's Det = 0

(4) $|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ then

$a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} = |A|$

$a_{11}C_{21} + a_{12}C_{22} + a_{13}C_{23} = 0$

⑤ Area of $\triangle ABC$ where $A(x_1,y_1)$, $B(x_2,y_2)$, $C(x_3,y_3)$ is $A=\frac{1}{2}\begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix}$

⑥ Condition for Collinearity is Area $=0$ i.e $\begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix}=0$

⑦ $$\frac{d}{dx}\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}=\begin{vmatrix} a' & b' & c' \\ d & e & f \\ g & h & i \end{vmatrix}+\begin{vmatrix} a & b & c \\ d' & e' & f' \\ g & h & i \end{vmatrix}+\begin{vmatrix} a & b & c \\ d & e & f \\ g' & h' & i' \end{vmatrix}$$

⑧ $|A|=|A^T|$, ⑨ $|A^m|=|A|^m$ for eg $|A^3|=|A|^3$

⑩ $|ABC|=|A|\cdot|B|\cdot|C|$ ⑪ $|A^{-1}|=\frac{1}{|A|}$ i.e $\det(A^{-1})=\frac{1}{(\det A)}$

$|\text{adj}\,A|=|\text{Cof}\,A|$ ⑫ if $A_{n\times n}$ then $|\text{adj}\,A|=|A|^{(n-1)}$

(13) A is called Singular if $|A| = 0$

& " " Non Singular if $|A| \neq 0$

(14) E-Operations →

(i) $R_i \leftrightarrow R_j$

(ii) $R_i \longrightarrow kR_i$

(iii) $R_i \longrightarrow R_i + kR_j$

for eg $R_3 \to R_3 - 2R_1$

(*) when we apply 3rd E-operation in a Mat. A, Value of it's det remains unaltered.

(*) If A is given Mat and after using $R_1 \leftrightarrow R_4$ Mat will become B then $|B| = -|A|$

(*) if A is given Mat & after using $R_2 \to 5R_2$ Mat will become B then

$$|B| = 5.|A|$$

Eg: $A=\begin{bmatrix}3 & -2 & 1\\ 0 & 4 & 5\\ 1 & 3 & -4\end{bmatrix}$ then verify prop (4)

Sol: $|A| = \cdots\cdots\cdots = -107$

$C_{11}=(-1)^2[-16-15]=-31$

$C_{12}=(-1)^3[0-5]=+5$

$C_{13}=(-1)^4[0-4]=-4$

$C_{21}=(-1)^3[8-3]=-5$

$C_{22}=(-1)^4[-12-1]=-13$

$C_{31}=(-1)^4[-10-4]=-14$

$C_{23}=(-1)^5(9+2)=-11$

$C_{32}=(-1)^5[15-0]=-15$

$C_{33}=(-1)^6[12-0]=12$

Now $a_{11}C_{11}+a_{12}C_{12}+a_{13}C_{13}=?$

$=3(-31)-2(5)+1(-4)=-93-10-4=-107$ ✓

Again $a_{12}C_{12}+a_{22}C_{22}+a_{32}C_{32}$

$=-2(5)+4(-13)+3(-15)=-10-52-45$

$=-107$ ✓

Now $a_{11}C_{31}+a_{12}C_{32}+a_{13}C_{33}$

$=3(-14)-2(-15)+1(12)$

$=-42+30+12=0$

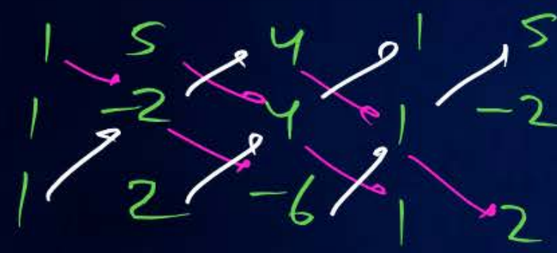
$\text{adj}A=\begin{bmatrix}-31 & 5 & -4\\ -5 & -13 & -11\\ -14 & -15 & 12\end{bmatrix}$

Q: Find the area of the $\triangle$ formed by $(5,4)$, $(-2,4)$, $(2,-6)$

Sol: $\text{Area} = \frac{1}{2}\begin{vmatrix} 1 & 5 & 4 \\ 1 & -2 & 4 \\ 1 & 2 & -6 \end{vmatrix} \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - R_1} = \frac{1}{2}\begin{vmatrix} 1 & 5 & 4 \\ 0 & -7 & 0 \\ 0 & -3 & -10 \end{vmatrix} = \frac{1}{2}\{1\{+70 - 0\}\}$

$= 35$ sq. units.

(M-II)



So $|A| = (12+20+8) - (-8+8-30) = 40+30 = 70$

So $\text{Area} = \frac{1}{2}|A| = \frac{1}{2}(70) = 35$

Q: Find λ if the points $(2,-3)$, $(\lambda,-1)$, $(0,4)$ are collinear?

Sol: Collinear $\Rightarrow$ Area $= 0 \Rightarrow \frac{1}{2}\begin{vmatrix} 1 & 2 & -3 \\ 1 & \lambda & -1 \\ 1 & 0 & 4 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} 1 & 2 & -3 \\ 1 & \lambda & -1 \\ 1 & 0 & 4 \end{vmatrix} = 0$

$\Rightarrow \lambda = 10/7$

Q: If $f(x)=\begin{vmatrix} x & x^2 & x^3 \\ 1 & 2 & 3x \\ 0 & 2 & 5x \end{vmatrix}$ s.t. $f'(x)=-1$ then $x=?$

Sol: $f'(x)=\begin{vmatrix} 1 & 2x & 3x^2 \\ 1 & 2 & 3x \\ 0 & 2 & 5x \end{vmatrix}+\begin{vmatrix} x & x^2 & x^3 \\ 0 & 0 & 3 \\ 0 & 2 & 5x \end{vmatrix}+\begin{vmatrix} x & x^2 & x^3 \\ 1 & 2 & 3x \\ 0 & 0 & 5 \end{vmatrix}$

$=\{1[10x-6x]-1[10x^2-6x^2]\}+\{-3[2x-0]\}+\{+5[2x-x^2]\}$

$=4x-4x^2-6x+10x-5x^2=-9x^2+8x$

Atq, $f'(x)=-1$

$-9x^2+8x=-1$

$9x^2-8x-1=0$

$9x^2-9x+x-1=0$

$(x-1)(9x+1)=0 \Rightarrow x=1 \text{ or } -\frac{1}{9}$

$A=\begin{bmatrix}3 & -2 & 1\\ 0 & 4 & 5\\ 1 & 3 & -4\end{bmatrix}$ then ① $|A|=?=-107$

② $|A^T|=?=\begin{vmatrix}3 & 0 & 1\\ -2 & 4 & 3\\ 1 & 5 & -4\end{vmatrix}=\ldots\ldots=-107$

③ $|A^2|=?=|A|^2=(-107)^2$ Ans

④ $|\bar{A}^1|=?=\frac{1}{|A|}=\frac{-1}{107}$ Ans

⑤ $|adj\,A|=?=|A|^{n-1}=(-107)^{3-1}$

Now $Cof\,A=\begin{bmatrix}-31 & 5 & -4\\ -5 & -13 & -11\\ -14 & -15 & 12\end{bmatrix}\Rightarrow adj\,A=(Cof\,A)^T=\begin{bmatrix}-31 & -5 & -14\\ 5 & -13 & -15\\ -4 & -11 & 12\end{bmatrix}\Rightarrow |adj\,A|=\ldots\ldots=(-107)^2$

Ex $A=\begin{bmatrix}3 & -2\\ -1 & 2\end{bmatrix}$ then $|A|, |A^3|, |\bar{A}^1|$

Sol: $|A|=6-(2)=4$

$|A^3|=|A|^3=(4)^3=64$

$|\bar{A}^1|=\frac{1}{|A|}=\frac{1}{4}$

(M-II) $\bar{A}^1=\frac{adj\,A}{|A|}=\frac{1}{4}\begin{bmatrix}2 & 2\\ 1 & 3\end{bmatrix}=\begin{bmatrix}1/2 & 1/2\\ 1/4 & 3/4\end{bmatrix}$

$|\bar{A}^1|=\frac{1}{2}\times\frac{3}{4}-\frac{1}{4}\times\frac{1}{2}=\frac{2}{8}=\frac{1}{4}$

Q. If $|A| = -1$, $|B| = 3$, $|C| = 5$ then $|(ABC)| = ?$ $= |A| \cdot |B| \cdot |C|$

$= (-1)(3)(5) = -15$

Q. Find x if $A = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 2 & 1 \\ x & 2 & -3 \end{bmatrix}$ is singular

" A is given as singular, $\Rightarrow |A| = 0$

$$\begin{vmatrix} 1 & -2 & 3 \\ 1 & 2 & 1 \\ x & 2 & -3 \end{vmatrix} = 0 \dots\dots\rightarrow \boxed{x = -1}$$

Q. $A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$ then $|A| = 11$

(i) Now $R_1 \leftrightarrow R_2$ then $B = \begin{bmatrix} 3 & 4 \\ 2 & -1 \end{bmatrix} \Rightarrow |B| = -11 = -|A|$

(ii) Now $R_2 \rightarrow 3R_2$

$C = \begin{bmatrix} 2 & -1 \\ 9 & 12 \end{bmatrix}$

$\Rightarrow |C| = 24 - (-9) = 33$

$= 3 \times 11$

$|C| = 3|A|$

Q: $|A| = \begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{vmatrix} = ?$ $\xrightarrow{C_1 \to C_1 + (C_2 + C_3 + C_4)}$ $= \begin{vmatrix} 10 & 2 & 3 & 4 \\ 10 & 3 & 4 & 1 \\ 10 & 4 & 1 & 2 \\ 10 & 1 & 2 & 3 \end{vmatrix}$

$= 10 \begin{vmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 4 & 1 \\ 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 3 \end{vmatrix}$ $\begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \\ R_4 \to R_4 - R_1 \end{matrix}$ $\longrightarrow 10 \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & -3 \\ 0 & 2 & -2 & -2 \\ 0 & -1 & -1 & 1 \end{vmatrix}$

$= 10 \begin{vmatrix} 1 & 1 & -3 \\ 2 & -2 & -2 \\ -1 & -1 & 1 \end{vmatrix}$ $\xrightarrow{R_3 \to R_3 + R_1}$ $10 \begin{vmatrix} 1 & 1 & -3 \\ 2 & -2 & -2 \\ 0 & 0 & -2 \end{vmatrix}$

$= (10)\left[(-2)\{-2-2\}\right] = 10(8) = 80$ //


Thank
YOU